

UniPrep Demo Computer Science Exit Exam

Sample 2 - Parser-Friendly MCQ PDF for Import, Review, Quiz, and Analytics Demonstration

Instructions: Choose the best answer for each question. Each item uses a simple format: question text, four options, and an explicit correct-answer line. This document is designed for UniPrep PDF-import testing.

Total Questions: 30 **Suggested Time:** 45 minutes

Programming and Algorithms

1. Which data structure follows the First-In, First-Out principle?

- A. Stack
- B. Queue
- C. Tree
- D. Graph

Correct Answer: B

2. What is the average-case time complexity of binary search on a sorted array?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$

Correct Answer: B

3. Which sorting algorithm repeatedly selects the smallest remaining element and places it in its correct position?

- A. Merge sort
- B. Quick sort
- C. Selection sort
- D. Heap sort

Correct Answer: C

4. In object-oriented programming, which concept allows one class to acquire properties and methods from another class?

- A. Encapsulation
- B. Inheritance
- C. Abstraction
- D. Overloading

Correct Answer: B

5. Which traversal visits the root node before its subtrees?

- A. Inorder
- B. Postorder
- C. Preorder
- D. Level order only

Correct Answer: C

6. What is the worst-case time complexity of linear search?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n^2)$

Correct Answer: C

System Development

7. Which SQL command is used to remove all rows from a table while keeping the table structure?

- A. DELETE DATABASE
- B. DROP TABLE
- C. TRUNCATE TABLE
- D. REMOVE ROWS

Correct Answer: C

8. Which normal form removes partial dependency on a composite key?

- A. First Normal Form
- B. Second Normal Form
- C. Third Normal Form
- D. Boyce-Codd Normal Form

Correct Answer: B

9. What does ACID stand for in database transactions?

- A. Atomicity, Consistency, Isolation, Durability
- B. Accuracy, Control, Integrity, Data
- C. Access, Consistency, Indexing, Distribution
- D. Atomicity, Concurrency, Integration, Dependency

Correct Answer: A

10. Which HTTP method is normally used to retrieve a resource without modifying it?

- A. POST
- B. PUT
- C. GET
- D. DELETE

Correct Answer: C

11. In software engineering, a use case primarily describes:

- A. Database indexing rules
- B. Interactions between an actor and the system
- C. Source code formatting
- D. Network routing tables

Correct Answer: B

12. Which software development model emphasizes short iterations and continuous customer feedback?

- A. Waterfall
- B. Agile
- C. Big Bang
- D. Code-and-fix only

Correct Answer: B

Computer Networking and Security

13. Which protocol is responsible for translating domain names into IP addresses?

- A. FTP
- B. DNS
- C. SMTP
- D. DHCP

Correct Answer: B

14. Which OSI layer is responsible for end-to-end transport and reliability?

- A. Physical
- B. Data Link
- C. Transport
- D. Presentation

Correct Answer: C

15. What is the default port number for HTTPS?

- A. 21
- B. 25
- C. 80
- D. 443

Correct Answer: D

16. Which security property ensures that information is not modified without authorization?

- A. Confidentiality
- B. Integrity
- C. Availability
- D. Non-repudiation only

Correct Answer: B

17. Which device forwards packets between different networks?

- A. Hub
- B. Switch
- C. Router
- D. Repeater

Correct Answer: C

18. Which attack attempts to make a service unavailable by overwhelming it with traffic?

- A. Phishing
- B. Denial-of-Service
- C. SQL normalization
- D. Digital signing

Correct Answer: B

Computer Architecture and Operating Systems

19. Which component performs arithmetic and logical operations in a processor?

- A. Control Unit
- B. ALU
- C. Cache controller
- D. I/O port

Correct Answer: B

20. Which scheduling algorithm gives each process a fixed time quantum?

- A. First Come First Served
- B. Shortest Job First
- C. Round Robin
- D. Priority without preemption

Correct Answer: C

21. A deadlock can occur when processes wait indefinitely for resources held by each other. Which condition is required for deadlock?

- A. Resource sharing
- B. Circular wait
- C. Fast execution
- D. Cache coherence

Correct Answer: B

22. Virtual memory allows a system to:

- A. Replace the CPU
- B. Use secondary storage as an extension of main memory
- C. Disable paging
- D. Remove all interrupts

Correct Answer: B

23. Which memory is closest to the CPU and generally fastest?

- A. Hard disk
- B. Main memory
- C. Cache memory
- D. Optical storage

Correct Answer: C

24. The main purpose of an operating system is to:

- A. Design hardware circuits
- B. Manage resources and provide services to programs
- C. Replace application software
- D. Convert all programs to assembly manually

Correct Answer: B

Compiler and Complexity

25. Which phase of a compiler converts a stream of characters into tokens?

- A. Code generation
- B. Lexical analysis
- C. Optimization
- D. Linking

Correct Answer: B

26. A deterministic finite automaton differs from a nondeterministic finite automaton because a DFA:

- A. Has exactly one transition for each symbol from a state
- B. Uses a stack
- C. Accepts only context-free languages
- D. Cannot have final states

Correct Answer: A

27. Which grammar type is commonly used to describe programming language syntax?

- A. Regular grammar only
- B. Context-free grammar
- C. Unrestricted grammar only
- D. No grammar is needed

Correct Answer: B

28. Which data structure is commonly used by a parser for nested constructs?

- A. Stack
- B. Queue
- C. Hash table only
- D. Array without indices

Correct Answer: A

29. A problem is in class P if it can be solved:

- A. Only by a nondeterministic machine
- B. In polynomial time by a deterministic algorithm
- C. Only using exponential memory
- D. Without any algorithm

Correct Answer: B

30. Which compiler phase checks type compatibility and declaration rules?

- A. Lexical analysis
- B. Semantic analysis
- C. Code generation
- D. Loading

Correct Answer: B

Answer Key

1. B 2. B 3. C 4. B 5. C 6. C 7. C 8. B 9. A 10. C

11. B 12. B 13. B 14. C 15. D 16. B 17. C 18. B 19. B 20. C

21. B 22. B 23. C 24. B 25. B 26. A 27. B 28. A 29. B 30. B

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